

The Languages of the Product-Mix Auctions

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joint work with Elizabeth Baldwin and Paul Klemperer

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UNIVERSITY OF OXFORD

After the Northern Rock bank run in 2007-8, Paul Klemperer designed the Product-Mix Auction (PMA) for the Bank of England. Still used to allocate funds to banks.

- The BoE's original PMA design used a simple geometric bidding language [Klemperer, 2008].
- Design has been extended in various papers [Klemperer, 2008, 2016] and studied algorithmically [Baldwin et al., 2022, 2023].
- PMA variant for the Central Bank of Iceland to buy back “offshore” accounts.

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- PMA variant for the Central Bank of Iceland to buy back “offshore” accounts.
- IMF staff interested in PMA variant to re-negotiate sovereign debt.
- Collaboration with industry partner to design PMA for trading Investment Tax Credits (ITCs).

PMA bidding languages: systematic way for participants to submit preferences before auction is run.

Provide theoretical underpinnings of PMA.

- Geometric approach to construct 'sensible' bidding languages from 'elementary' building blocks.
- Leads to family of bidding languages for the Product-Mix Auctions (PMA).

New contribution

1. Representations theorems:

- *all substitutes* preferences (and no other preferences) can be expressed in the *substitutes PMA language*.
- *all strong-substitutes* preferences (and no other preferences) can be expressed in the *strong-substitutes PMA language*.

2. New way of understanding substitutes and strong-substitutes.

3. Implies algorithms for learning an agent's preferences in these languages.

SSRN preprint:



Other bidding languages

- ‘Assignment messages’ [Milgrom, 2009] can’t express all strong-substitutes prefs. [Fichtl, 2021].
- Family of ‘logical languages’ by [Nisan, 2006], [Lehmann, Lehmann and Nisan, 2006] similar in spirit but not geometric. Don’t capture strong-substitutes and substitutes classes.
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(Strong) substitutes

- Substitutes widely studied in matching, housing, labour markets.
- Strong-substitutes guarantees comp. equilibrium existence in multi-unit settings [Milgrom & Strulovici, 2009].
- Deep connections to discrete convex analysis ($M^{\natural}$ -concavity) [Shioura & Tamura, 2015].

Product-Mix Auction trading **bundles** $x \in \mathbb{Z}^n$ of **indivisible goods** $[n] := \{1, \dots, n\}$ available in **multiple units**.

Each agent has:

- Concave **valuation function** $v : A \rightarrow \mathbb{R}$ defined on finite set A of bundles.
- **Quasilinear utility** $v(x) - \mathbf{p} \cdot x$ for bundle x at prices $\mathbf{p} \in \mathbb{R}^n$.
- **Demand** $D_v(\mathbf{p}) := \arg \max_{x \in A} (v(x) - \mathbf{p} \cdot x)$.

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Definition (Kelso and Crawford, 1982)

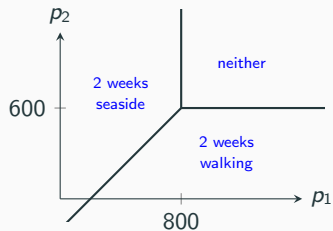
Valuation v is *substitutes* if increasing prices of some goods doesn't reduce demand in the goods with unchanged prices.

It is *strong-substitutes* if it is substitutes when treating each unit as a separate good with its own price.

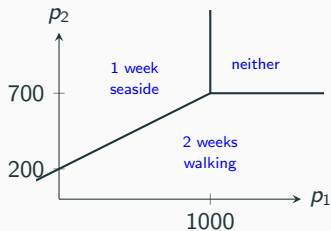
Geometric characterisation of (strong) substitutes later.

Examples

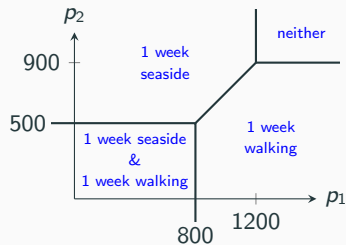
Three buyers for two kinds of holidays: **seaside** (good 1) and **walking** (good 2).



Antigone



Bathsheba



Calpurnia

The Product-Mix Bidding Languages

In the PMA, every auction participant submits a collection $\mathcal{B}$ of *bids*.

- Each bid defines an 'elementary' demand.
- The participant's demand is the 'aggregate' of the bids' demands.
- More general 'elementary' bids lead to more expressive PMA language.

The Dot Bid language (Bank of England)

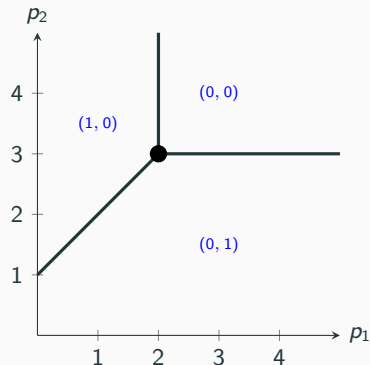
A single **dot bid** b at root $r \in \mathbb{R}_{\geq 0}^n$ expresses *unit demand*: “want at most one item.”

Vector r defines valuation

$$v(x) = \begin{cases} r_i & \text{if } x = e^i, \\ 0 & \text{if } x = \mathbf{0}, \\ -\infty & \text{else.} \end{cases}$$

At prices p :

- Bid demands *good* i maximising *utility* $r_i - p_i > 0$.
- Demands *nothing* if utilities of all goods are negative.
- In case of indifferences, break ties arbitrarily.



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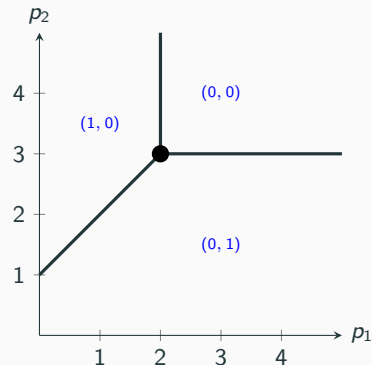
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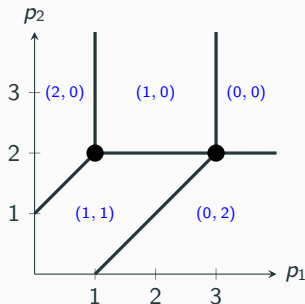
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The bid's **demand (correspondence)** is $D_b(p) := \{e^i \mid i \in J_b(p)\}$ with $J_b(p) = \arg \max_{i \in [n]} (r_i - p_i)$.

The Dot Bid language (Bank of England)

Defining the demand $D_{\mathcal{B}}(\mathbf{p})$ of a collection $\mathcal{B}$ of dot bids:



- At generic prices, $D_{\mathcal{B}}(\mathbf{p})$ is the sum of the unique bundles demanded by the bids.

$$D_{\mathcal{B}}(\mathbf{p}) := \sum_{b \in \mathcal{B}} D_b(\mathbf{p}).$$

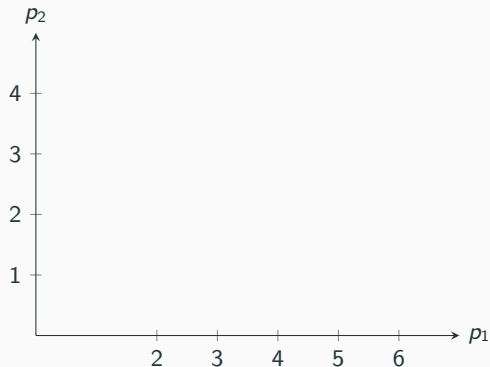
- At indifference prices, take the discrete convex hull of the aggr. demand at generic prices $Q(\mathbf{p})$ in the neighbourhood:

$$D_{\mathcal{B}}(\mathbf{p}) := \mathbb{Z}^n \cap \text{conv}\{D_{\mathcal{B}}(\mathbf{q}) \mid \mathbf{q} \in Q(\mathbf{p})\}.$$

Generalisation: bids $\mathbf{b} = (\mathbf{r}, m)$ with *multiplicity* $m > 0$: represent m ‘unit demand’ dot bids with same root $\mathbf{r}$.

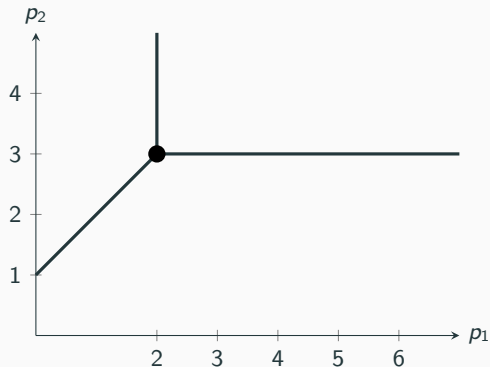
The PMA language with negative bids

So far, we can express any aggregation of unit demands.



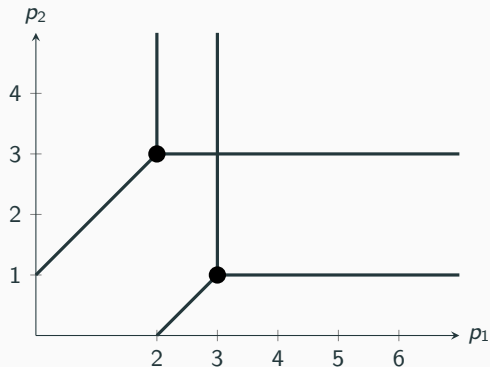
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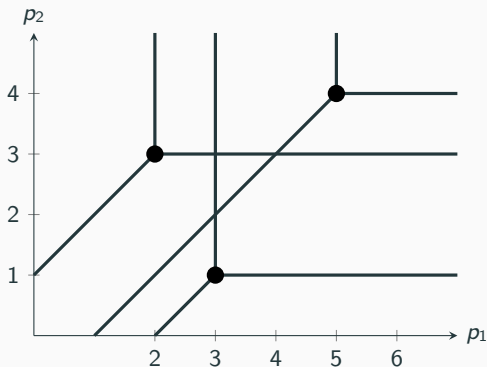
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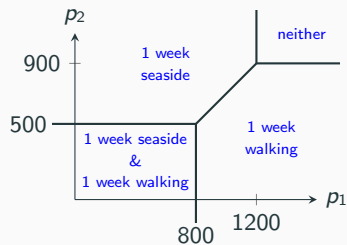
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Easy for bidders and auctioneer. Similar to OXS 'logical language' [Lehmann et al., 2006].

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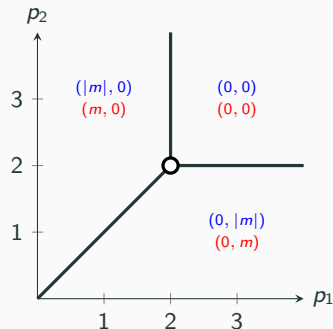
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But can't represent Calpurnia's preferences!

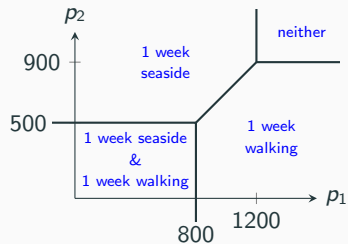
Understanding negative bids

Extension: allow **negative bids** (r, m) with $m < 0$.

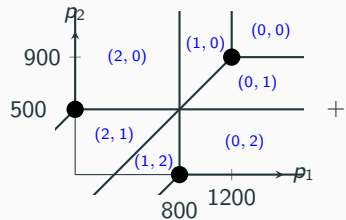
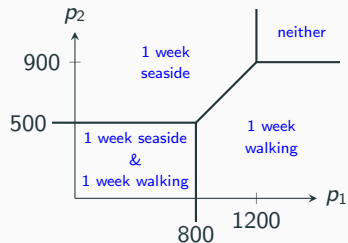


- Demand of negative bid (r, m) is negation of bid $(r, |m|)$'s demand.
- Negative bids 'remove' units from aggregation $D_B(\mathbf{p})$.
- NB: Demanding negative units is *not* the same as selling!

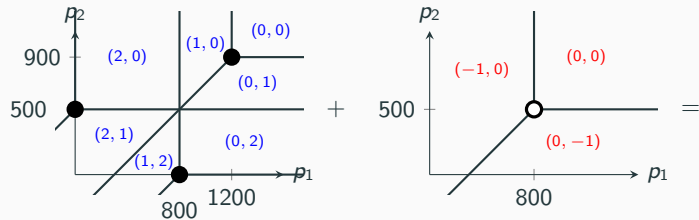
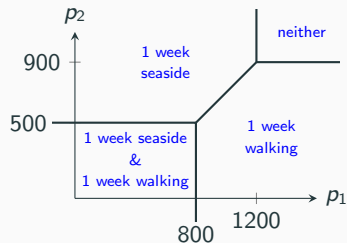
Calpurnia's demand



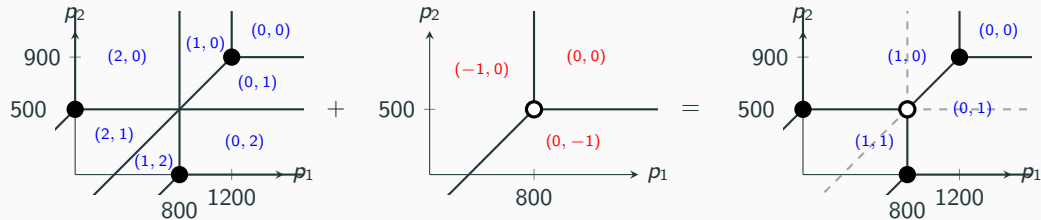
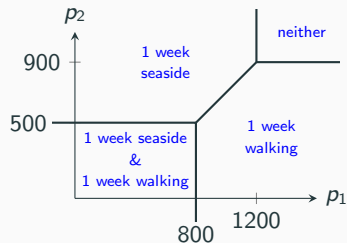
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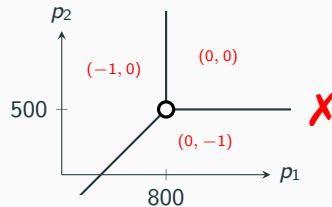
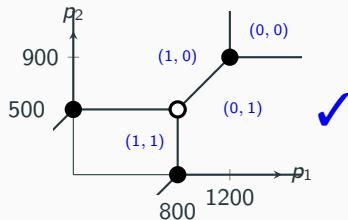
Geometrically, negative bids 'cancel out' indifference price lines.

Valid bid collections

PMA participants submit bid collections that express the demand of a valuation.

Definition

A bid collection $\mathcal{B}$ is *valid* if there exists a valuation v with $D_{\mathcal{B}} = D_v$.

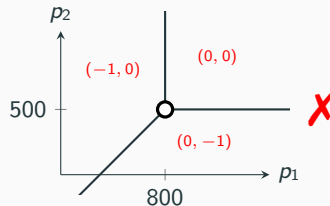
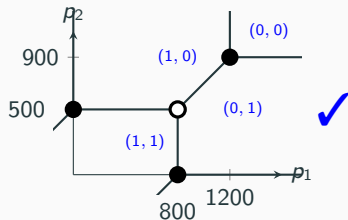


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Characterisation: $\mathcal{B}$ is valid iff $D_{\mathcal{B}}$ satisfies the *law of demand*: increasing the price of one good (weakly) reduces the demand for that good.

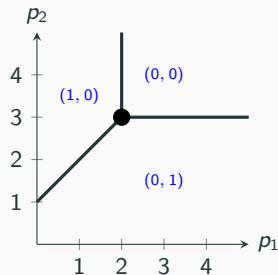
Paper also develops various algorithmically checkable characterisations of validity.

Regular preferences

Dot bid language can capture any strong-substitutes preferences satisfying natural 'regularity' property:

Definition: Demand D is *regular* if, for all goods $i \in [n]$:

1. good i is not demanded if its price is high enough,
2. after reducing the price of good i sufficiently, while keeping all other prices fixed, only good i is demanded.



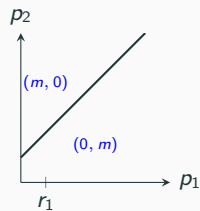
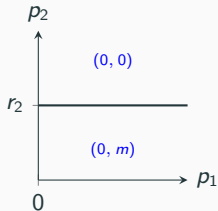
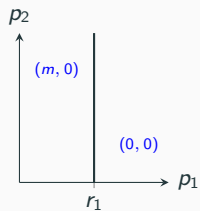
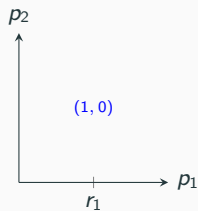
Representation theorem (regular version)

Every *regular* strong-substitutes valuation v has a collection $\mathcal{B}$ of dot bids with $D_{\mathcal{B}} = D_v$.

Proof sketch later.

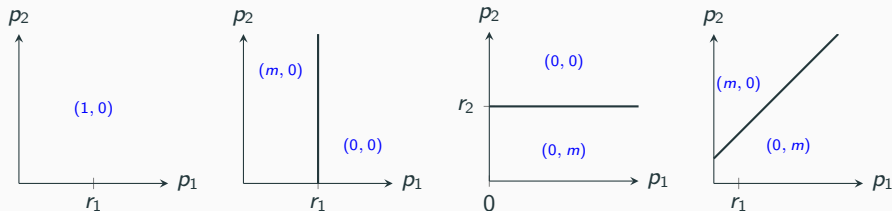
Non-regular building blocks

Non-regular 'elementary' strong-substitutes preferences:



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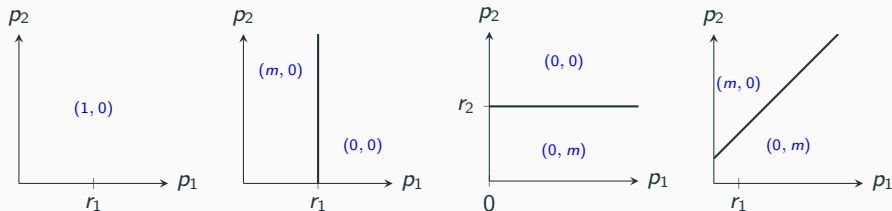


Strong-substitutes PMA language introduces non-regular bids (r, m) interested only in some goods.

- Allow root entry $r_i = -\infty$ if not interested in good $i \in \{0, \dots, n\}$ at any price.
- Utility of such goods will be $-\infty$. (Definition of bid demand D_b unchanged.)

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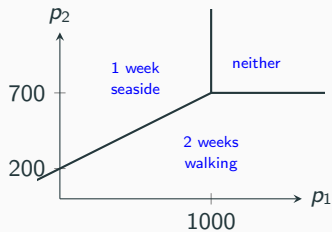
Representation theorem (second part)

Every strong-substitutes valuation v has a strong-substitutes PMA bid collection $\mathcal{B}$ with $D_{\mathcal{B}} = D_v$.

Non-regular bids important for sellers and traders on both sides of market. Less pertinent for PMA.

Substitutes PMA language

'Elementary' either-or demands with different quantities. E.g., "I want two apples or one banana."

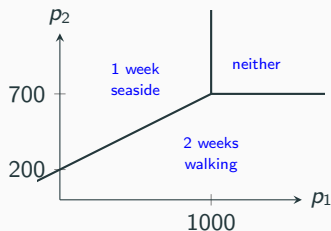


Bathsheba's bid:
 $r = (1000, 700),$

$$m = 1.$$

Substitutes PMA language

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Bathsheba's bid:

$$r = (1000, 700),$$

$$t = (1, 2),$$

$$m = 1.$$

In **substitutes PMA language**, bids $b = (r; t; m)$ have additional **tradeoff vector** t capturing relative tradeoffs between good quantities.

- At generic prices, bid demands quantity mt_i of utility-maximizing good i .
- Otherwise, bid demands any integer bundle in $\text{conv}\{mt_i \mid i \in J\}$ of utility-maximising goods J .

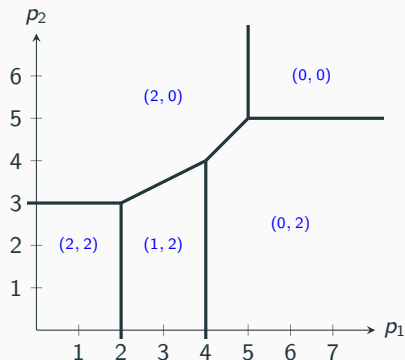
Representation Theorem (first part)

Every substitutes valuation v has a substitutes PMA bid collection $\mathcal{B}$ with $D_{\mathcal{B}} = D_v$.

The Geometry of Demand

Material from [Baldwin and Klemperer, 2019].

Locus of Indifferences Prices (LIP) $\mathcal{L}_v := \{\mathbf{p} \in \mathbb{R}^n \mid |D_v(\mathbf{p})| \geq 2\}$ for valuation v .

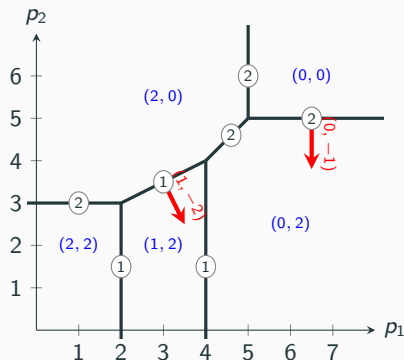


- $\mathcal{L}_v$ decomposes into $n - 1$ -dimensional linear pieces, or **facets**.
- Divides price-space into *unique demand regions (UDRs)*.

The LIP

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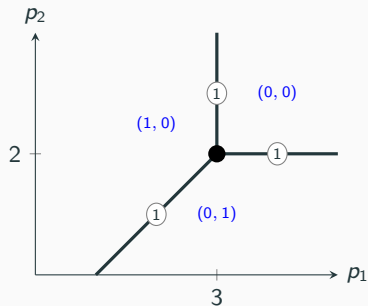
- $\mathcal{L}_v$ decomposes into $n - 1$ -dimensional linear pieces, or **facets**.
- Divides price-space into *unique demand regions (UDRs)*.
- The (primitive) normal vectors $\mathbf{n}$ of each facet capture *relative* change in demand.
- Give each facet F a weight $w_v(F) > 0$ so that demand change in direction $\mathbf{n}$ is $-w_v(F)\mathbf{n}$.

Fact: The **weighted LIP** fully specifies demand changes of D_v .

We can also define weighted LIPs for bid collections $\mathcal{B}$.

First, $\mathcal{L}_{\mathcal{B}} := \{\mathbf{p} \in \mathbb{R}^n \mid |D_{\mathcal{B}}(\mathbf{p})| \geq 2\}$.

Fact: For an individual (strong) substitutes bid $\mathbf{b} = (\mathbf{r}; \mathbf{t}; m)$, we can write out the (strong) substitutes valuation $v_{\mathbf{b}}$ for which $D_{\mathbf{b}} = D_{v_{\mathbf{b}}}$.

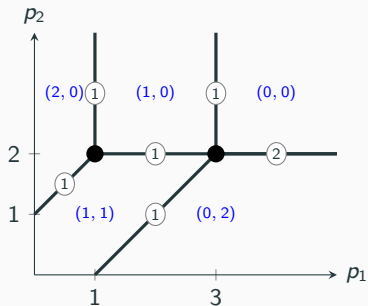


- Associate each individual bid $\mathbf{b} \in \mathcal{B}$ with weighted LIP $(\mathcal{L}_{\mathbf{b}}, w_{\mathbf{b}}) = (\mathcal{L}_{v_{\mathbf{b}}}, w_{v_{\mathbf{b}}})$.

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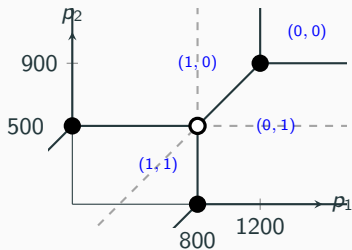


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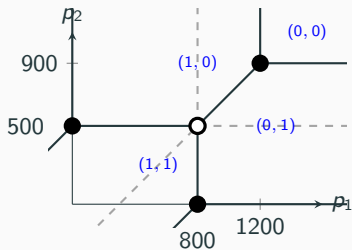


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- The weighted LIP $(\mathcal{L}_{\mathcal{B}}, w_{\mathcal{B}})$ is the 'sum' of the weighted bid LIPs.
- When negative bids are involved, we remove any facets with weight 0 that are 'cancelled' out.

Fact: Demand changes of $D_{\mathcal{B}}$ are fully described by the **weighted LIP** $(\mathcal{L}_{\mathcal{B}}, w_{\mathcal{B}})$.

From [Baldwin & Klemperer, 2019] (characterisation of demand types):

- Valuation v for two goods is *substitutes* iff every diagonal facets has positive slope.
- Valuation v for two goods is *strong-substitutes* if every diagonal facets has slope 1.

The LIPs of (strong) substitutes bid collections have positive slope (slope 1) by construction, so the corresponding valuation is (strong) substitutes.

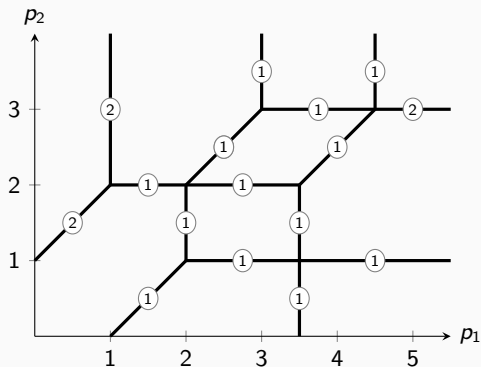
Proposition

The substitutes bidding languages only express substitutes preferences.

The strong substitutes bidding languages only express strong-substitutes preferences.

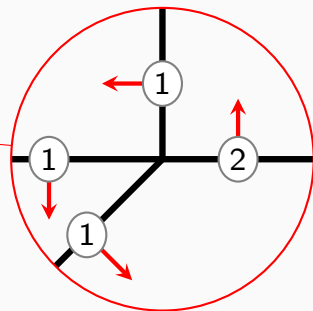
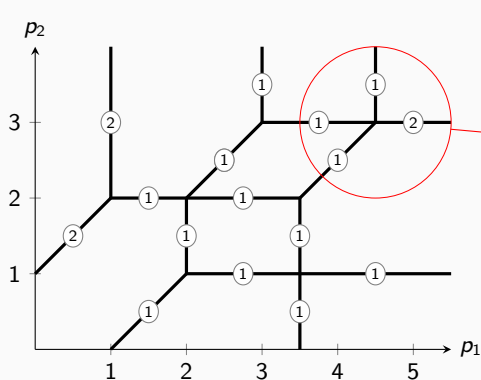
Geometric implications

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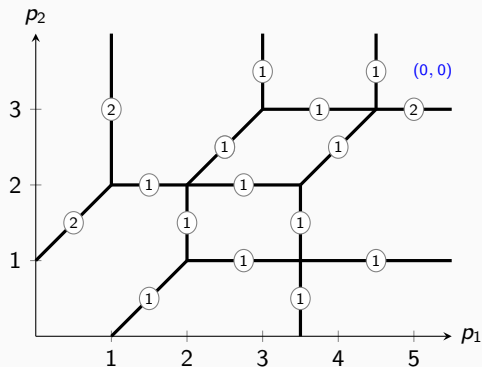
$$2 \cdot \begin{pmatrix} 0 \\ 1 \end{pmatrix} + \begin{pmatrix} -1 \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ -1 \end{pmatrix} + \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

Fact [Mikhalkin, 2004]

Weighted LIPs satisfy the 'balancing condition'.

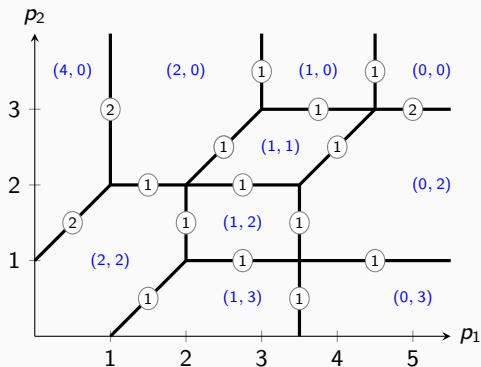
Geometric implications

Suppose $(\mathcal{L}, w)$ is a weighted LIP (of valuation v or bid collection $\mathcal{B}$).



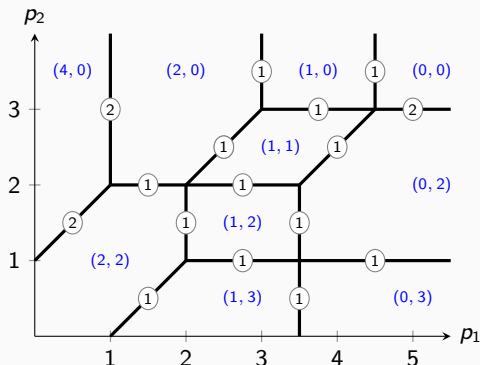
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Suppose $(\mathcal{L}, w)$ is a weighted LIP (of valuation v or bid collection $\mathcal{B}$).



Geometric implications

Suppose $(\mathcal{L}, w)$ is a weighted LIP (of valuation v or bid collection $\mathcal{B}$).



Proposition

If $(\mathcal{L}_v, w_v) = (\mathcal{L}_B, w_B)$ and $D_v(\mathbf{p}) = D_B(\mathbf{p})$ for some price $\mathbf{p}$, then $D_v = D_B$.

Proof Sketch of Representation Theorems

Representation Theorem

Every substitutes valuation v has a 'unique' substitutes bid collection $\mathcal{B}$ with $D_{\mathcal{B}} = D_v$.

Every strong-substitutes valuation v has a 'unique' strong-substitutes bid collection $\mathcal{B}$ with $D_{\mathcal{B}} = D_v$.

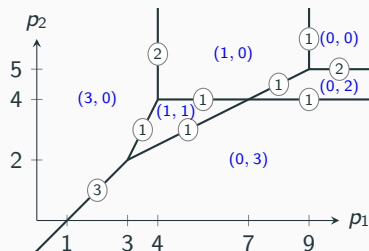
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High-level proof idea:

1. Devise procedure to build bid collection $\mathcal{B}$ from $(\mathcal{L}_v, w_v)$.
2. Prove that $D_v = D_{\mathcal{B}}$.



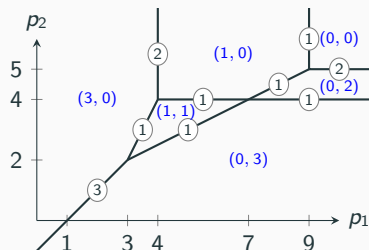
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High-level proof idea:

1. Devise procedure to build bid collection $\mathcal{B}$ from $(\mathcal{L}_v, w_v)$.
2. Prove that $D_v = D_{\mathcal{B}}$.
 - 2.1 Establish $(\mathcal{L}_{\mathcal{B}}, w_{\mathcal{B}}) = (\mathcal{L}_v, w_v)$.
 - 2.2 Show that $D_v(\mathbf{p}) = D_{\mathcal{B}}(\mathbf{p})$ for some generic price $\mathbf{p}$.
 - 2.3 This implies $D_v = D_{\mathcal{B}}$.



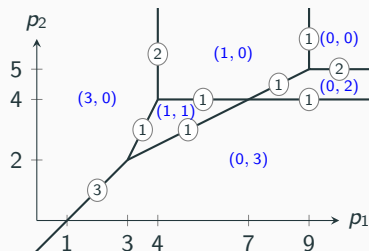
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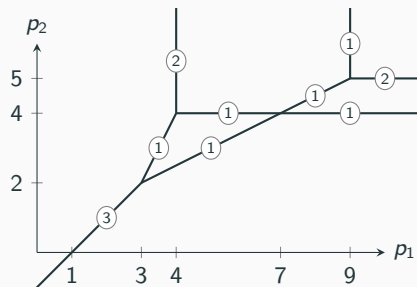
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 - 2.2 Show that $D_v(\mathbf{p}) = D_{\mathcal{B}}(\mathbf{p})$ for some generic price $\mathbf{p}$.
 - 2.3 This implies $D_v = D_{\mathcal{B}}$.
3. Argue that $\mathcal{B}$ contains only strong-substitutes bids if v is strong-substitutes.
4. Show that $\mathcal{B}$ is unique.



Constructing bid collection $\mathcal{B}$

For simplicity, let's assume v is regular.

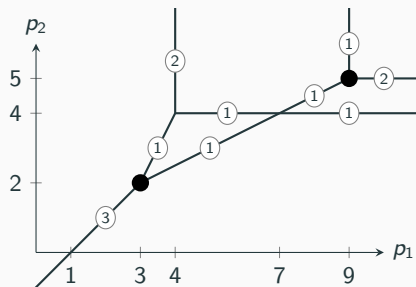


Constructing bid collection $\mathcal{B}$

For simplicity, let's assume v is regular.

Let $\Delta_v(\mathbf{p}; \frac{t_1}{t_2})$ be the difference in weight between the diagonal facets of slope $\frac{t_1}{t_2}$ below and above vertex $\mathbf{p}$ of weighted LIP $(\mathcal{L}_v, w_v)$. *Missing facets have weight 0!*

Ex: $\Delta_v((9, 5), \frac{1}{2}) = 1$ and $\Delta_v((3, 2), \frac{1}{1}) = 3$.



Constructing bid collection $\mathcal{B}$

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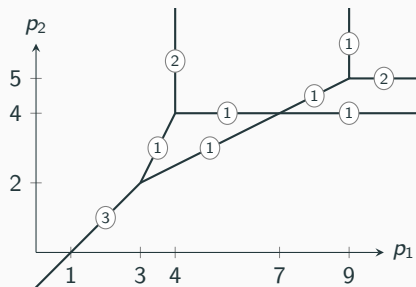
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Procedure for constructing $\mathcal{B}$

For every $v \times \mathbf{p}$ and slope $\frac{t_1}{t_2}$ of a diagonal facet meeting $\mathbf{p}$:

If $\Delta_v(\mathbf{p}; \frac{t_1}{t_2}) \neq 0$, add a regular bid with root $\mathbf{p}$, tradeoffs (t_1, t_2) and multiplicity $\Delta_v(\mathbf{p}; \frac{t_1}{t_2})$.



Constructing bid collection $\mathcal{B}$

For simplicity, let's assume v is regular.

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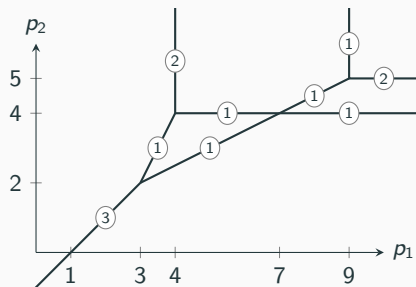
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Procedure for constructing $\mathcal{B}$

For every vx $\mathbf{p}$ and slope $\frac{t_1}{t_2}$ of a diagonal facet meeting $\mathbf{p}$:

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Bid	r	t	m
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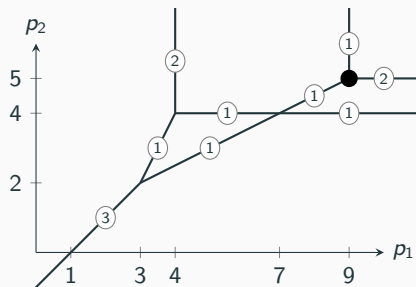
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Bid	r	t	m
1	(9, 5)	(1, 2)	1



Constructing bid collection $\mathcal{B}$

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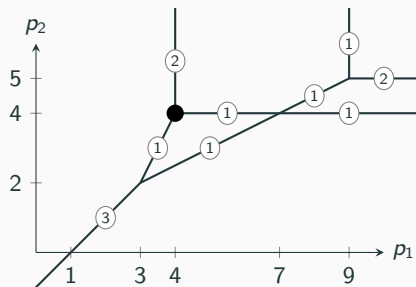
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1	(9, 5)	(1, 2)	1
2	(4, 4)	(2, 1)	1



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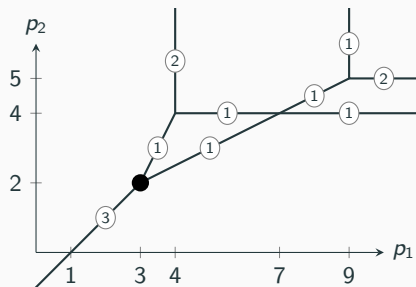
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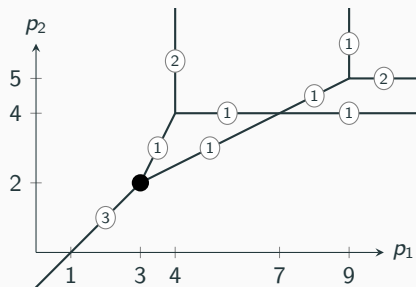
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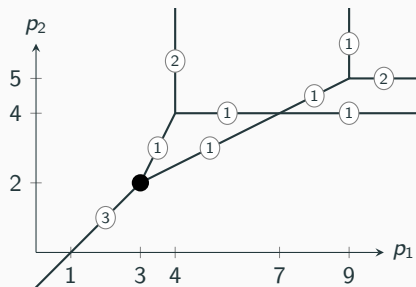
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2	(4, 4)	(2, 1)	1
3	(3, 2)	(1, 2)	-1
4	(3, 2)	(2, 1)	-1
5	(3, 2)	(1, 1)	3



Constructing bid collection $\mathcal{B}$

For simplicity, let's assume v is regular.

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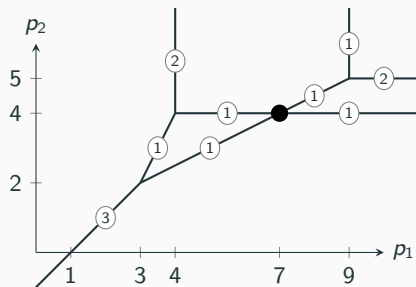
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5	(3, 2)	(1, 1)	3



Proving correctness

We argue that $(\mathcal{L}_v, w_v) = (\mathcal{L}_B, w_B)$.

Define Δ_b and Δ_B analogously to Δ_v for weighted LIPs $(\mathcal{L}_b, w_b)$ and $(\mathcal{L}_B, w_B)$.

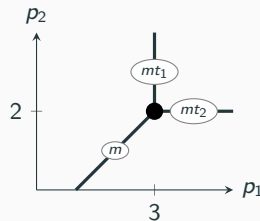
Difference Lemma

In both weighted LIPs, $\Delta_v(\mathbf{p}; \frac{t_1}{t_2}) = \Delta_B(\mathbf{p}; \frac{t_1}{t_2})$ for any point $\mathbf{p}$ and slope $\frac{t_1}{t_2}$.

Fix $\mathbf{p}$. By construction of B , $\Delta_v(\mathbf{p}; \frac{t_1}{t_2}) =: m \neq 0$ iff B contains a bid $(\mathbf{p}; \mathbf{t}; m)$.

We verify that $\Delta_b(\mathbf{p}; \frac{t_1}{t_2}) = m$. All other bids $\mathbf{b}'$ satisfy $\Delta_{\mathbf{b}'}(\mathbf{p}; \frac{t_1}{t_2}) = 0$. So

$\Delta_B(\mathbf{p}; \frac{t_1}{t_2}) = \Delta_b(\mathbf{p}; \frac{t_1}{t_2}) = m$.



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In both weighted LIPs, $\Delta_v(\mathbf{p}; \frac{t_1}{t_2}) = \Delta_B(\mathbf{p}; \frac{t_1}{t_2})$ for any point $\mathbf{p}$ and slope $\frac{t_1}{t_2}$.

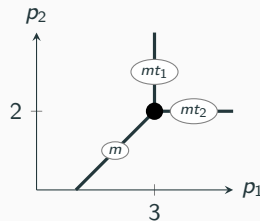
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$\Delta_B(\mathbf{p}; \frac{t_1}{t_2}) = \Delta_b(\mathbf{p}; \frac{t_1}{t_2}) = m$.

Lemma

$(\mathcal{L}_v, w_v)$ and $(\mathcal{L}_B, w_B)$ have the same weighted diagonal facets.



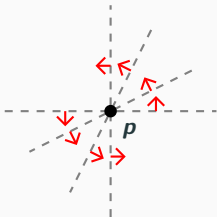
- Fix a doubly-infinite line through any diagonal facet of $\mathcal{L}_v$ or $\mathcal{L}_B$.
- Any sufficiently high price on the line lies in the UDR demanding $\mathbf{0}$ due to regularity.
- By Difference Lemma, the weights of facets change in both wLIPs in the same way.
- So both have the same weighted facets along the line.

Proving correctness

Next we show that both weighted LIPs also have same horizontal and vertical facets.

Key Lemma

At any prices $\mathbf{p}$, the diagonal facet weight differences $\Delta(\mathbf{p}; \frac{t_1}{t_2})$ determine the weight difference $\Delta^{hori}(\mathbf{p})$ between left and right horizontal facets, and the difference $\Delta^{vert}(\mathbf{p})$ between lower and upper vertical facets.



Balancing condition:

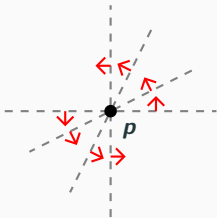
$$\sum_{\frac{t_1}{t_2}} \Delta(\mathbf{p}; \frac{t_1}{t_2}) \begin{pmatrix} t_1 \\ t_2 \end{pmatrix} + \Delta^{hori} \begin{pmatrix} 0 \\ -1 \end{pmatrix} + \Delta^{vert} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}.$$

Proving correctness

Next we show that both weighted LIPs also have same horizontal and vertical facets.

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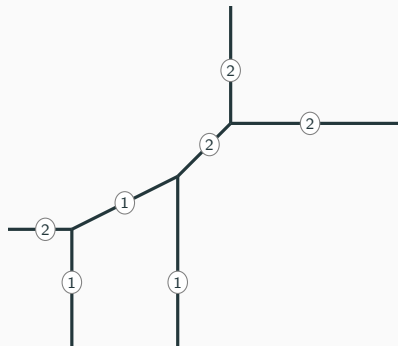
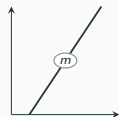
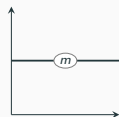
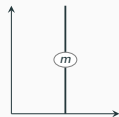
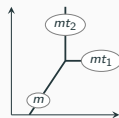
Balancing condition:

$$\sum_{\frac{t_1}{t_2}} \Delta(\mathbf{p}; \frac{t_1}{t_2}) \begin{pmatrix} t_1 \\ t_2 \end{pmatrix} + \Delta^{hori} \begin{pmatrix} 0 \\ -1 \end{pmatrix} + \Delta^{vert} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}.$$

Now apply same inductive argument as for diagonal lines:

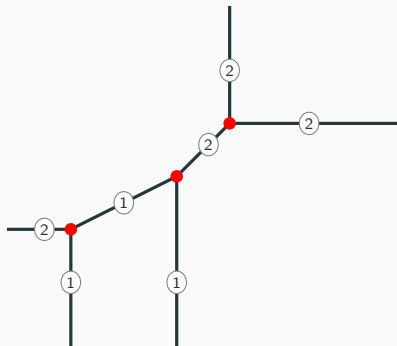
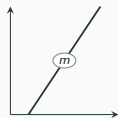
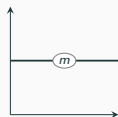
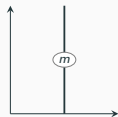
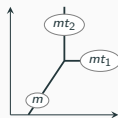
- Fix any horizontal (vertical) line in $\mathcal{H}_v$ or $\mathcal{H}_B$.
- Due to regularity, any sufficiently low price on the line lies inside a UDR (not in a facet).

Non-regular valuations



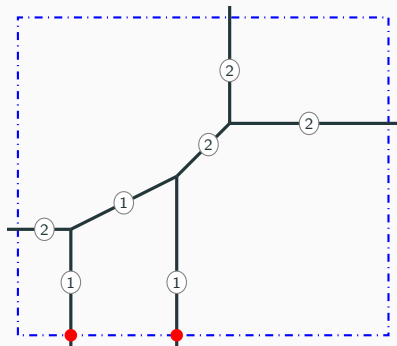
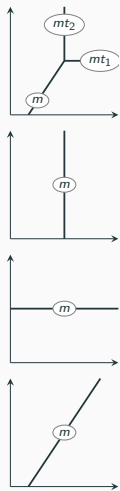
1. For every v_x of $\mathcal{L}_V$:
add **type 1** bids with matching slope and multiplicity as before.
2. For every intersection of lower boundary with vertical line:
add **type 2** bid with m equal to adjacent facet weight.
3. Similarly, add **type 3** bids for intersections of left boundary with horizontal line.
4. For every intersection of upper or right boundary with diagonal line:
add **type 4** bid with matching slope and m equal to adjacent facet weight.

Non-regular valuations



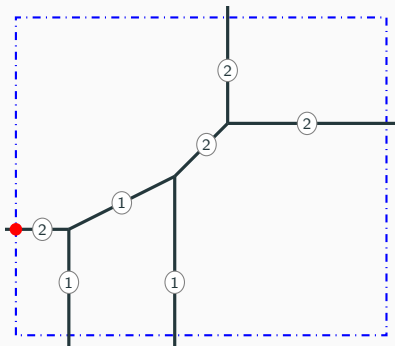
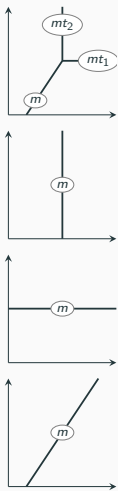
1. For every vx of $\mathcal{L}_v$:
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Non-regular valuations



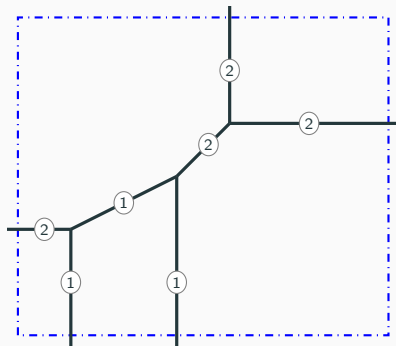
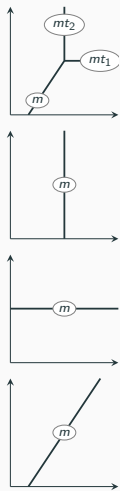
1. For every vx of $\mathcal{L}_v$:
add **type 1** bids with matching slope and multiplicity as before.
2. For every intersection of lower boundary with vertical line:
add **type 2** bid with m equal to adjacent facet weight.
3. Similarly, add **type 3** bids for intersections of left boundary with horizontal line.
4. For every intersection of upper or right boundary with diagonal line:
add **type 4** bid with matching slope and m equal to adjacent facet weight.

Non-regular valuations



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Main takeaways

- PMA bidding languages express substitutes and strong-substitutes preference class.
- New way of thinking about these classes.
- Imply algorithms for eliciting preferences from PMA participants using demand queries.
- Algorithmically tractable for auctioneer to compute competitive equilibrium (cf. [BGKL, 2023]).

Thank you

SSRN preprint:

